

'AGP 3.0'. This is a 32-bit-wide parallel bus clocked at a whopping 533 MHz, offering a total bandwidth of about 2.1 GBps, doubling the currently available 1 GBps provided by AGP 4x. Sadly, AGP 8x isn't truly backwards compatible and in its current implementation doesn't support either AGP 1x or 2x signals, meaning older AGP 1x or 2x cards simply won't function in an AGP 8x system.

Super storage

By the beginning of the coming year you'll find 200 GB drives as widespread as you find 40 GB drives today. We are already approaching the technical limitations of the current ATA standard; today's controllers cannot recognise drives larger than 137.4 GB as they use a 28-bit addressing system. New standards will adopt a 48-bit addressing system, raising the maximum drive capacity a million fold to 144 petabytes (a petabyte equals a million gigabytes). But some drive manufacturers want to take storage to almost unrealistic heights. IBM is using a new disk material, nicknamed 'pixie dust'. Placed between two layers of magnetic material, it helps data remain stable at very high densities and IBM is aiming for 500 GB to 600 GB four-platter desktop drives by 2003!

The only way to support such large and fast drives will be to transition from the existing Parallel ATA standard to Serial ATA (SATA). Serial ATA starts at a data transfer rate of 150 MBps and will soon migrate to 300 MBps and then to a whopping 600 MBps. It uses a thinner, more flexible cable that improves airflow and thereby facilitates the development of smaller form factor systems with better thermal designs. Also, 5,400-rpm drives will start their slow

and inevitable march into hard drive history, while 7,200-rpm drives will become the standard and will ensure that the SATA interface will be pushed to its limits. By the end of this year you will see third-party chipset solutions supporting SATA and by the beginning of next year SATA will be integrated into most mainstream motherboard chipsets.

Perfect port ability

It's time to throw away your 700 MB CD-Rs! The price of DVD-Rs has dropped as low as Rs 300 per disc and DVD-Writers have already started appearing in the market. Beginning next year DVD-Writers will cost as low as Rs 15,000 and DVD-Rs as a storage medium will start replacing CD-Rs.

The best part is that by the middle of next year we'll see the adoption of DVD's successor—the Blu-ray disc. This 12-cm disc will be the same size as its DVD counterpart but will have a maximum capacity of 27 GB per side (compared with 4.7 GB for DVD discs) and a data transfer rate of 36 MBps to accommodate high-definition content recording. The format would allow a disc to store more than 2 hours of High Definition video content and 13 hours of standard television broadcasts! To top it all, a double layer disc with up to 50 GB of storage capacity is also in the pipeline.

The final nail in the coffin for floppies is being hammered in by the new USB micro drives. These tiny Flash drives fit in the palm

of your hand and store up to 1 GB of data! Just plug it into a USB port and begin using it. These drives work seamlessly with any operating system, be it Macintosh, Linux or Windows and they don't even need any special device drivers.

Idiot-proof interfaces

In a year's time the PC will no longer support legacy I/O ports. You will not find any PS/2, serial or parallel ports residing on the back plate of your PC. These will be replaced by high-speed USB 2.0 and IEEE-1394 (FireWire) ports that are nearly 40 times faster than USB 1.0 and over 500 times faster than parallel ports. You will no longer face problems interfacing cutting-edge digicams, MP3 players, digital video recorders and external storage devices with your PC. Scanners and printers too will join the faster USB revolution and will shed their old and rather slow parallel interfaces, allowing for faster

ILLUSTRATION: Ram

On the desk

Delightful displays

Thin Film Transistor (TFT) LCDs are here to stay but currently they lack the knockout punch to throw out CRT monitors. That knockout punch can only be delivered by price. Today, you can buy a decent 15-inch TFT screen for about Rs 35,000. This is way beyond most people's budgets, but with TFT manufacturing yields improving rapidly, by 2003, you will be able to buy a 15-inch TFT for as low as Rs 20,000 and a 17-inch TFT for around Rs 30,000!

It's a no-brainer that in one year these

displays will become quite widespread and more and more PCs will be built using them. In the coming year, TFTs will truly come of age and will feature ultra fast response times. This means that there will be no smear or blurring in fast moving images, and games and movies will look crystal clear. Also, as the technology improves, it will solve most of the angle viewing limitations of today's TFTs and the next generation displays won't experience any image or contrast degradation when viewed from any angle.

Worry-free wiring

True liberation and freedom from the clutter of wires that plagues all computer users will occur only when the PC sheds off its rather tangled backside. One of the few areas where the PC has advanced considerably has been the speedy introduction (and adoption) of wireless input devices. Wireless keyboards have taken us to the next level of typing freedom. These are relatively expensive and out of reach for most users, but with more manufacturers jumping onto this bandwagon, with newer and more innovative designs